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定义势能函数

$$\Phi(D_i) = t(D_i) + \alpha L(D_i)$$

$$\text{其中 } \alpha = \frac{2}{k-1}$$

$$L(D_i) = \sum_{x \in V} S(x)$$

$S(x)$ 为结点 x 已失去的孩子数

当 $k=2$ 时得到 $\Phi(D_i) = t(D_i) + 2m(D_i)$ 标准形式

Decrease-key: 设共切掉了 h 个结点

$$\hat{c}_i = c_i + \Phi(D_i) - \Phi(D_{i-1})$$

$$= \Theta(h) + \Delta t + \alpha \Delta L$$

$$= \Theta(h) + h + \alpha \sum_{i=1}^h (-s(V_i) + 1)$$

$$= \Theta(h) + h + \frac{2}{k-1} \left[h - (s_1 + \sum_{i=2}^h k) \right] \rightarrow h - s_1 - (h-1)k$$

$$= \Theta(h) - h + \frac{2}{k-1} (k - s_1)$$

$$= \frac{2}{k-1} (k - s_1)$$

$$\leq \frac{2k}{k-1}$$

$$\left\{ \begin{array}{l} k=2 \text{ 时 } \hat{c}_i = 4 \\ k>2 \text{ 时 } \frac{2k}{k-1} < 4 \end{array} \right\} \text{ 降低了常数倍证明}$$

Delete-Min: 主要是 consolidate 代价 $\uparrow$

看 $D(n)$, $k=2$ 时, $\deg(Y_i) \geq i-k = i-2$

$$\text{size}(x) \geq S_k = 1 + \text{size}(Y_1) + \dots + \text{size}(Y_m)$$

$$\geq 2 + S_0 + \dots + S_{m-2} = \Theta\left(\left(\frac{1+\sqrt{5}}{2}\right)^m\right)$$

而 $k>2$ 时 $\deg(Y_i) \geq i-k$

$$\therefore \text{size}(x) \geq S_k \geq c + \sum_{i=k}^m S_{i-k} = \Theta(k^m) \quad k < \frac{1+\sqrt{5}}{2}$$

$$\therefore D(n) \leq \lg_k n \leq \lg_{\frac{1+\sqrt{5}}{2}} n$$

$$\therefore \text{提升了 } \frac{\lg_k n}{\lg_{\frac{1+\sqrt{5}}{2}} n} = \log_k \frac{1+\sqrt{5}}{2} \text{ 倍}$$

